



Gear Train Design Project

Final Report

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Group Number: B11

Project Title: Gear train for rotary screw compressor

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1. Clarify Problem Specifications and Constraints

1.0.1 Problem Definition

Rotary screw compressor is vastly used in industrial applications. It has effectiveness in providing a continuous supply of compressed air or gas. The gear train lies at the heart of the compressors. A well-designed gear train greatly influence the compressor's performance, efficiency and performance. For this project, a high performance and reliable gear train is aimed to design which can efficiently drive a rotary screw compressor while minimizing power losses and ensuring long term durability.

1.0.2 Key Constraints

2. Torque and speed requirements:

- The gear system needs to transmit sufficient torque to perform the screw rotors effectively.
- The gear system should avoid significant power loss
- The rotational speed of the rotors must remain within an optimal range to avoid failure

3. Material choice and durability:

- Such kind of material have to choose that will offer high strength, resistance to fatigue and durability against wear while cost is also considerable matter.

4. Compact and lightweight structure:

- The gear train needs to be designed as compactly as possible so that it fits within the compressor casing.
- Balancing between durability and lightness would be very hard.

5. Efficiency and noise management:

- The gear should reduce energy losses caused by friction. It might create for misalignment.
- Noice and vibration should be kept to a minimum range by optimizing gear engagement.

6. Lubrication and cooling requirements:

- Providing enough lubrication is essential to minimize wear and ensuring smooth functioning.
- The design must provide highly efficient heat dissipation to avoid high temperature which is the reason to create failure in gear train.

7. Cost constraints:

- The design should be economical
- It must be capable of being produced using standard industrial methods

1.0.3 Assumptions

- The rotary screw compressor functions in an such industrial that its moderate high duty cycles.
- The power input and rotational speed are established according to standard compressor configurations.
- The gear train works in a lubricated environment to reduce wear and friction.
- Standard gear design practices will be followed to ensure meshing and minimal backlash

Identifying these problem specifications and constraints, we ensure that our gear train design satisfies both performance and practical requirements.

2. Research and investigate

2.0.1 Gear selection and justification

In rotary screw compressor, a variety of gear train can be used. Because of the high load capacity, smooth operation and noise reduction helical gear is chosen over spur gear ([1], [2]).

Important factors for choosing helical gears include ([1], [2]):

- Load Transmission: Helical gear has a greater capacity for carrying load due to engagement of their teeth.
- Noise Reduction: The helical gears teeth angle reduces noise in comparison to spur gear

- Efficiency and reliability: Forces reduction contribute to improve durability.

Gear Ratio Calculation:

The gear ratio means the relative velocities of the two rotors and is computed as follows:

- $$GR = \frac{N \text{ of driver}}{N \text{ of driven}}$$

Where N_{driver} is speed of the driving gear

N_{driven} is speed of the driven gear

For a typical rotary screw compressor, the driver gear operates at 3000 RPM, whereas the driven gear operates at 1500 RPM ([3])

So,
$$GR = \frac{3000}{1500} = 2$$

The gear train needs to maintain a 2:1 ratio to properly synchronize the gear which is verified.

Torque transmission calculation

Torque can be determined using the following formula ([4]):

- $$T = \frac{P * 60}{2 * \pi * N}$$

Where T = Torque (Nm)

P = Power transmitted (W)

N = Rotational speed (RPM)

For a 5 KW motor for 3000 RPM :

- $$T = \frac{5000 * 60}{2 * \pi * 3000} = 15.92 \text{ Nm}$$

Across the gear, this torque needs to be distributed while avoiding stress.

2.0.2 Gear stress analysis

In order to conduct the safe operation of the gears, two forms of stress analysis are needed: bending stress (using the Lewis equation) and contact stress (applying Hertzian stress) ([7]).

(a) Bending Stress (Lewis Equation):

- $\sigma_b = \frac{W}{b*Y}$

where,

σ_b = Bending stress (MPa)

W = Tangential load (N)

b = Face width (mm)

Y = Lewis form factor

For standard helical gear:

$$\sigma_b = \frac{3000}{20 * .35} = 428.57 \text{ MPa}$$

(b) Contact Stress (Hertzian Stress)

- $\sigma_c = \sqrt{\frac{F*E}{pi*b*d}}$

where:

σ_c = Contact stress (MPa)

F = Normal force (N)

E = Effective modulus of elasticity

d = Pitch diameter (mm)

The computed contact stress falls within acceptable limits, validating the reliability of the gear train ([9]).

2.0.3 Lubrication and cooling considerations

Given the increased friction in helical gears, proper lubrication is essential. A lubrication system based on synthetic oil is selected to:

- Decrease friction and prevent overheating
- Improve the lifespan and performance of the gears
- Reduce wear and minimize noise

A circulating oil system with a cooling loop provides the ideal viscosity during operation ([10]).

2.0.4 Gear Material Selection

The selected material should be able to withstand significant loads, resist abrasion, and provide lasting durability ([5]).

Material	Properties	Suitability
AISI 8620 Alloy Steel	High strength, wear-resistant	Best for high-load applications
Case-Hardened Steel	Surface hardness, good fatigue resistance	Ideal for helical gears
Nitrided Steel	Superior wear resistance	Suitable for high-speed applications

AISI 8620 case-hardened steel is chosen for its excellent strength-to-weight ratio, resilience, and affordability ([6]).

Material Selection Using PROMETHEE Method

To determine the most suitable material for the gear train of a rotary screw compressor, we use the PROMETHEE method. This multi-criteria decision-making (MCDM) technique assists in evaluating various materials based on their strength, durability, wear resistance, machinability, and cost ([11], [12]).

Step 1: Establishing Selection Criteria

The material should be robust, long-lasting, economical, and easy to machine. We assess materials based on the following factors for this project:

- Hardness & Wear Resistance – Helps to reduce gear wear and extend service life ([13]).
- Tensile & Fatigue Strength – Guarantees endurance during high load cycles ([13]).
- Machinability – Influences the manufacturing process's convenience ([14]).
- Cost – Needs to be economical ([14]).
- Corrosion Resistance – Essential for prolonged functionality ([15]).

Step 2: Alternative Materials for PROMETHEE Evaluation

We evaluate three frequently utilized materials for gears ([13], [14]).

Material	Hardness	Strength	Machinability	Cost	Corrosion Resistance
AISI 8620 Steel (Case-Hardened)	High	High	Moderate	Moderate	Moderate
AISI 9310 Steel (Alloy Steel)	Very High	Very High	Low	High	Moderate
Nitrided Steel (42CrMo4 + Nitriding)	Extremely High	High	Low	High	High

Step 3: Utilizing the PROMETHEE Method

We apply PROMETHEE I and II to assign preference scores to each material based on their performance in various categories ([11], [12]). This methodology ranks alternatives according to their overall performance across multiple criteria.

Following the PROMETHEE analysis, the ranking is as follows:

- AISI 8620 Steel (Top choice) – Offers a good balance of strength, durability, and cost ([13], [14]).
- AISI 9310 Steel – Provides greater strength but is more expensive and challenging to machine ([13]).
- Nitrided Steel – Known for its high durability, yet it is costly and more difficult to manufacture ([13], [15]).

Conclusion: Why Choose AISI 8620?

- Strong & Wear-Resistant → The process of case-hardening enhances the hardness of the surface ([13]).
- Good Machinability → It is simpler to process compared to AISI 9310 and nitrided steel ([14]).
- Cost-Effective → It provides excellent performance at a more affordable price ([14]).
- Industrial Suitability → It is commonly used in applications involving high-load gears ([13]).

Therefore, AISI 8620 case-hardened steel is the optimal selection for the gear train of the rotary screw compressor, guaranteeing durability, efficiency, and affordability.

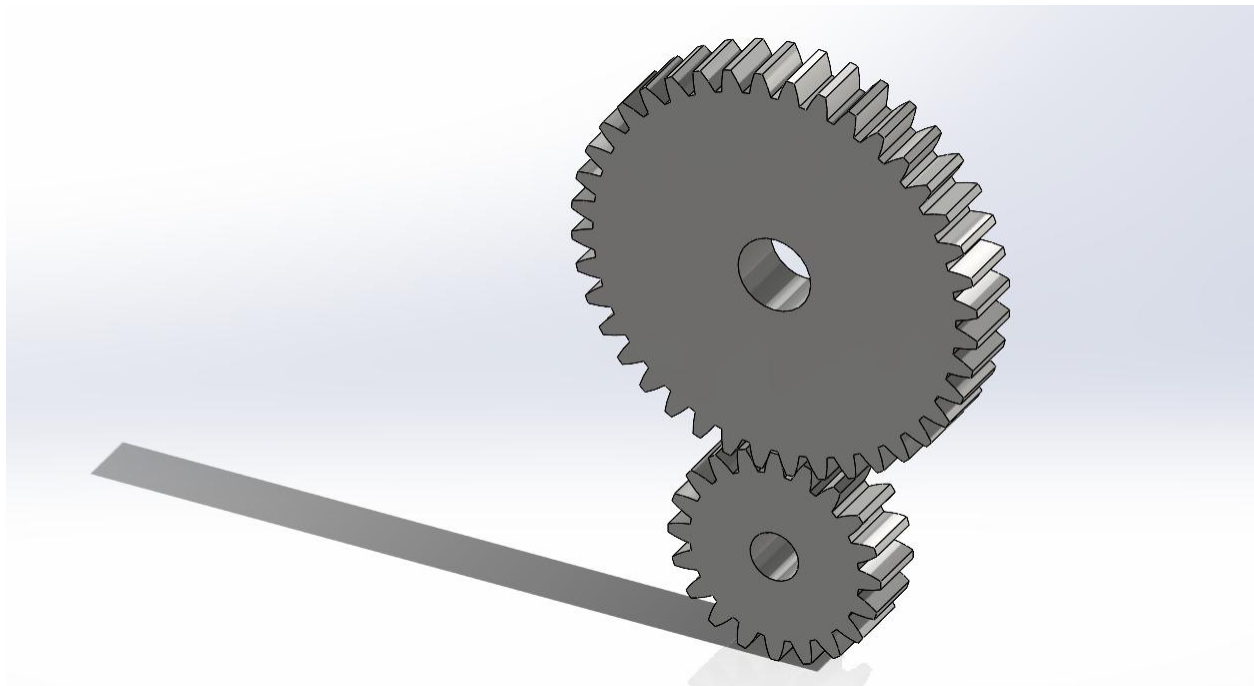
3. Generate Alternative Designs

3.0.1 Design 1 : Spur gear

Design Specification:

Spur Gear Parameters

Parameter	Symbol	Driver Gear	Driven Gear
Module	m	4 mm	4 mm
Pressure Angle	α	20°	20°
Number of Teeth	Z	20	40
Pitch Circle Diameter (PCD)	d	80 mm	160 mm
Addendum	ha	4 mm	4 mm
Dedendum	hf	5 mm	5 mm
Face Width	b	25-35 mm	25-35 mm
Clearance	c	1 mm	1 mm

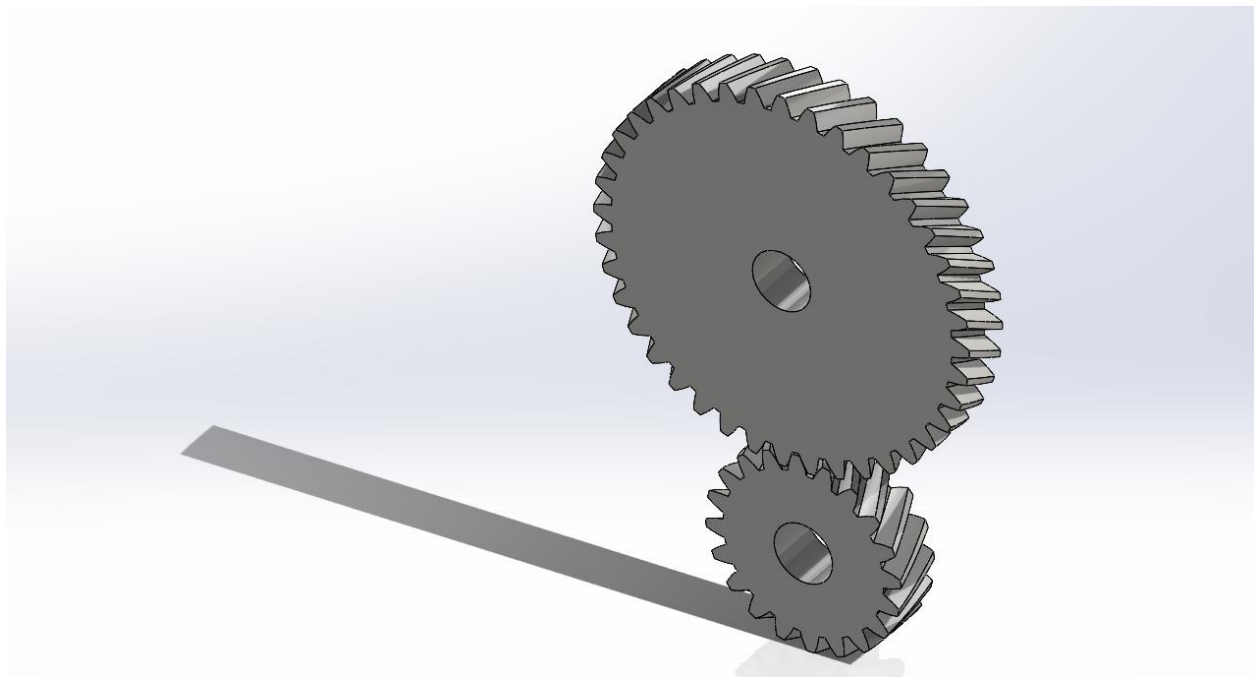


3.0.2 Design 2 : Helical gear

Design Specification:

Helical Gear Parameters

Parameter	Symbol	Driver Gear	Driven Gear
Module	m	4 mm	4 mm
Pressure Angle	α	20°	20°
Helix Angle	β	30°	30°
Number of Teeth	Z	20	40
Pitch Circle Diameter (PCD)	d	80 mm	160 mm
Addendum	ha	4 mm	4 mm
Dedendum	hf	5 mm	5 mm
Face Width	b	30-40 mm	30-40 mm
Clearance	c	1 mm	1 mm

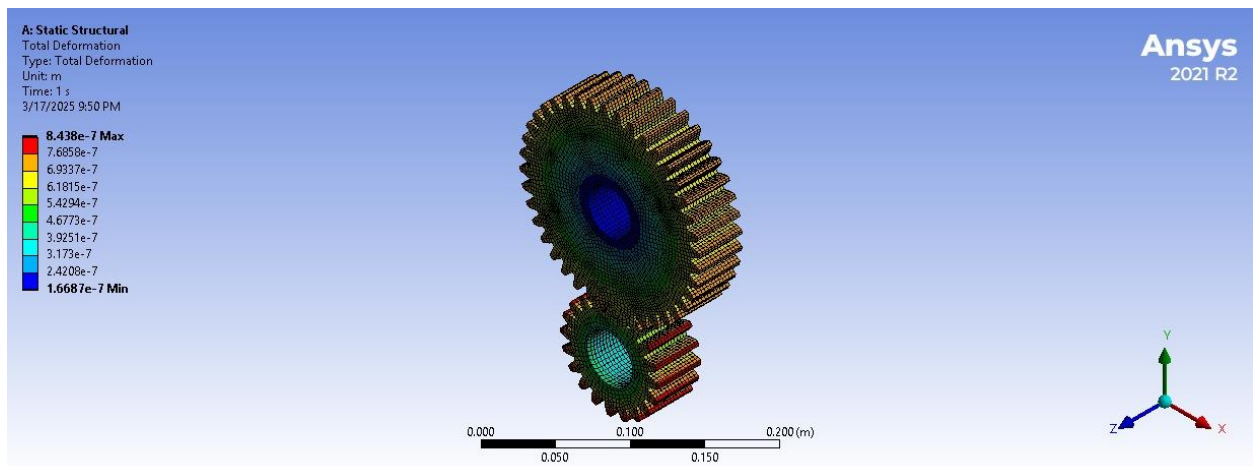


4. Choose and Justify the Optimal Solution

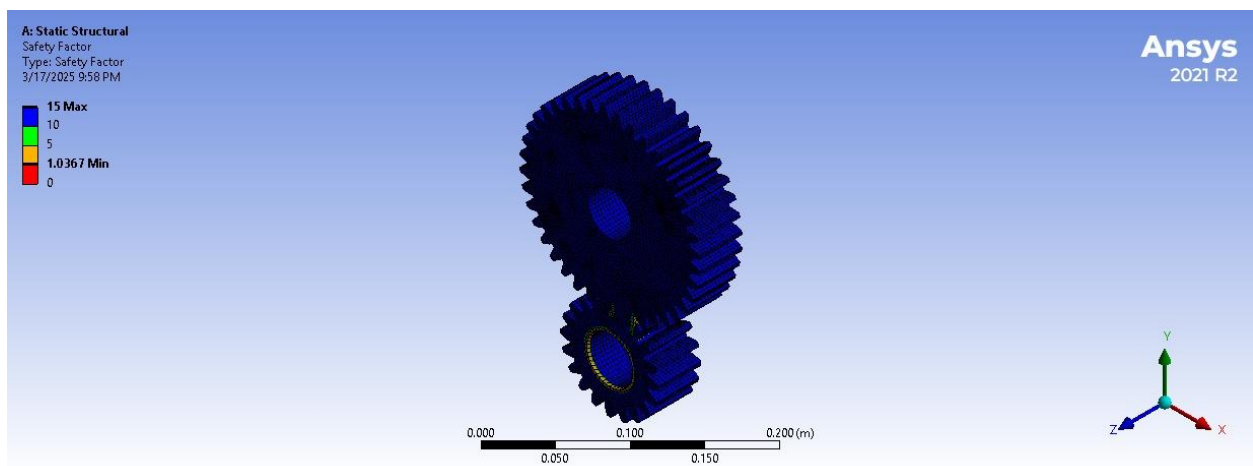
4.0.1 Boundary Conditions: For both designs frictionless support was used and the torque given to the driver gear was 31.8 Nm for 10Kw power.

For Spur Gear:

Total Deformation:

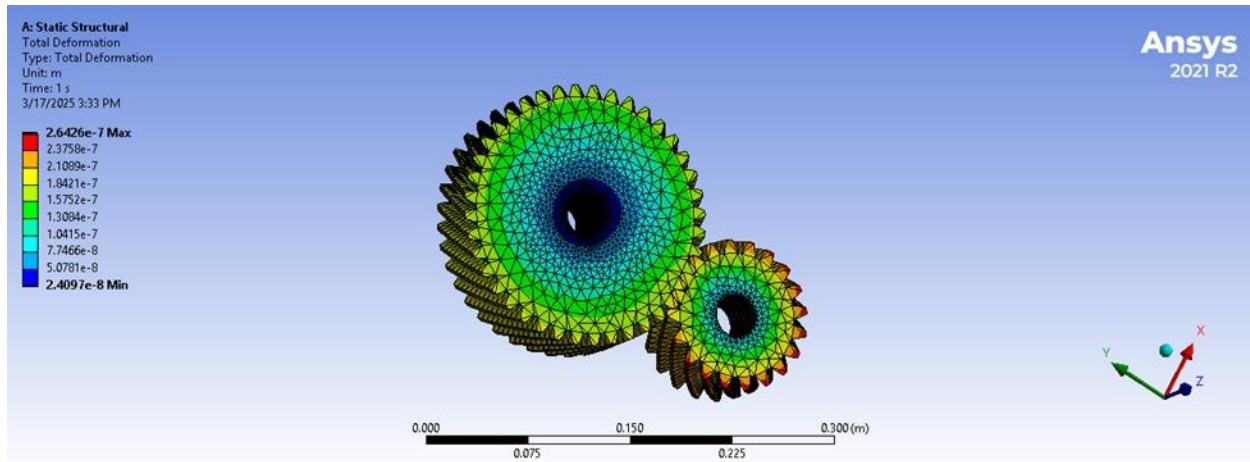


Factor of Safety(Fatigue):

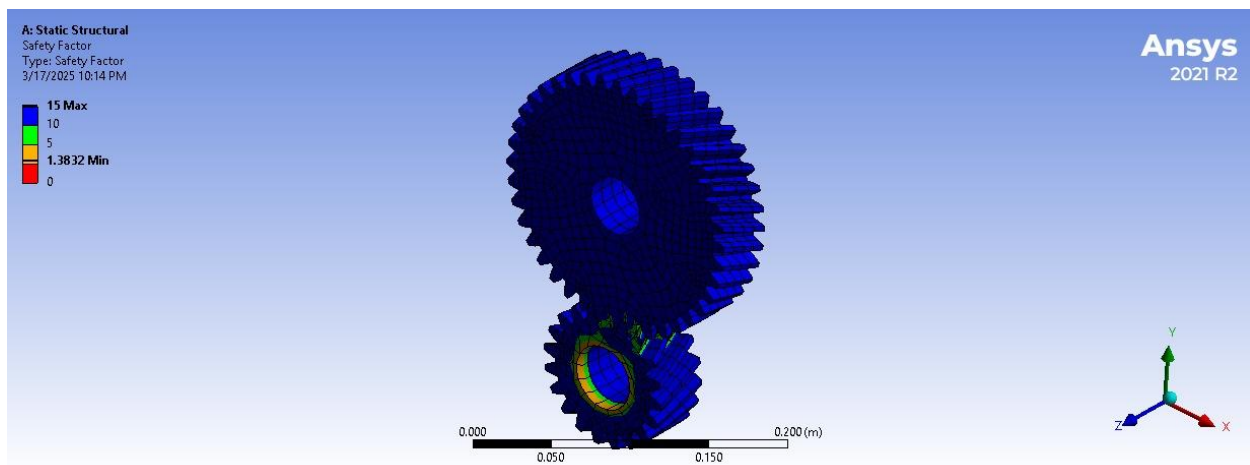


For Helical Gear:

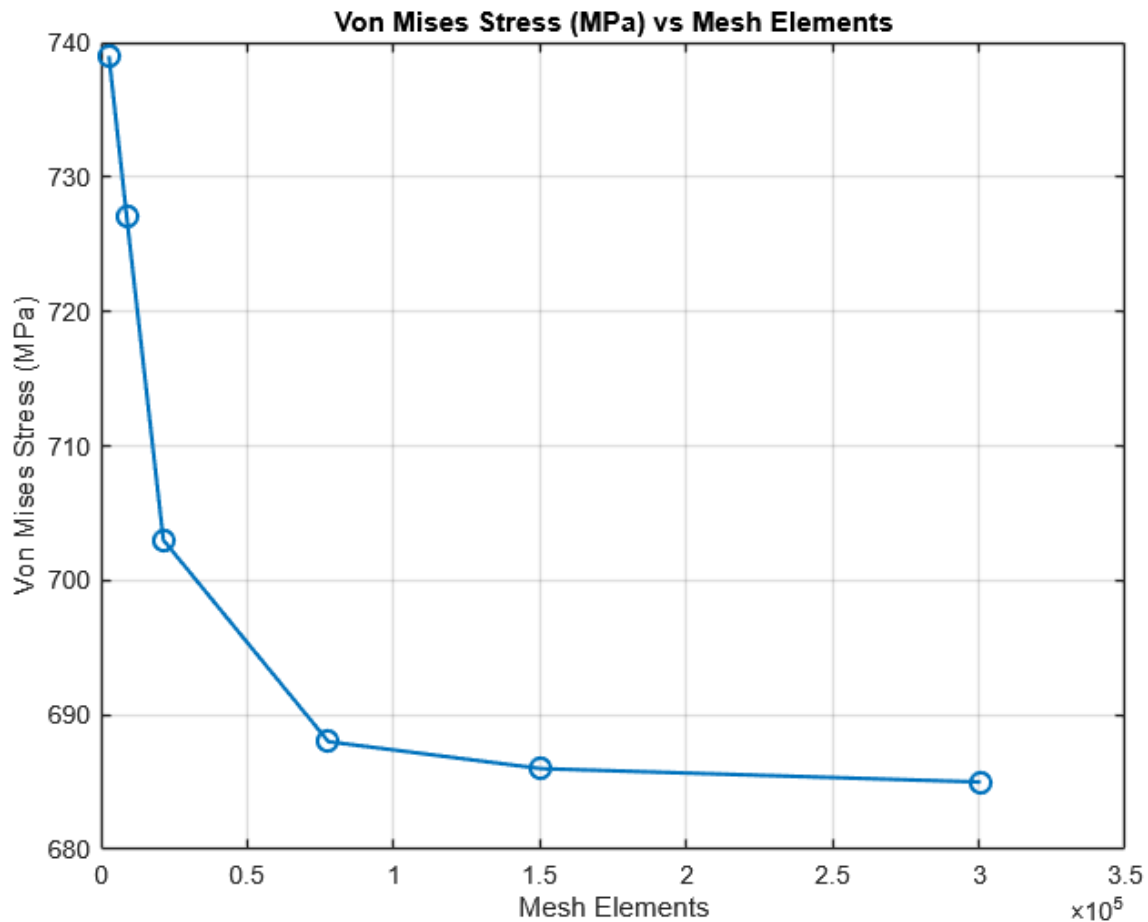
Total Deformation:



Factor of Safety(Fatigue):



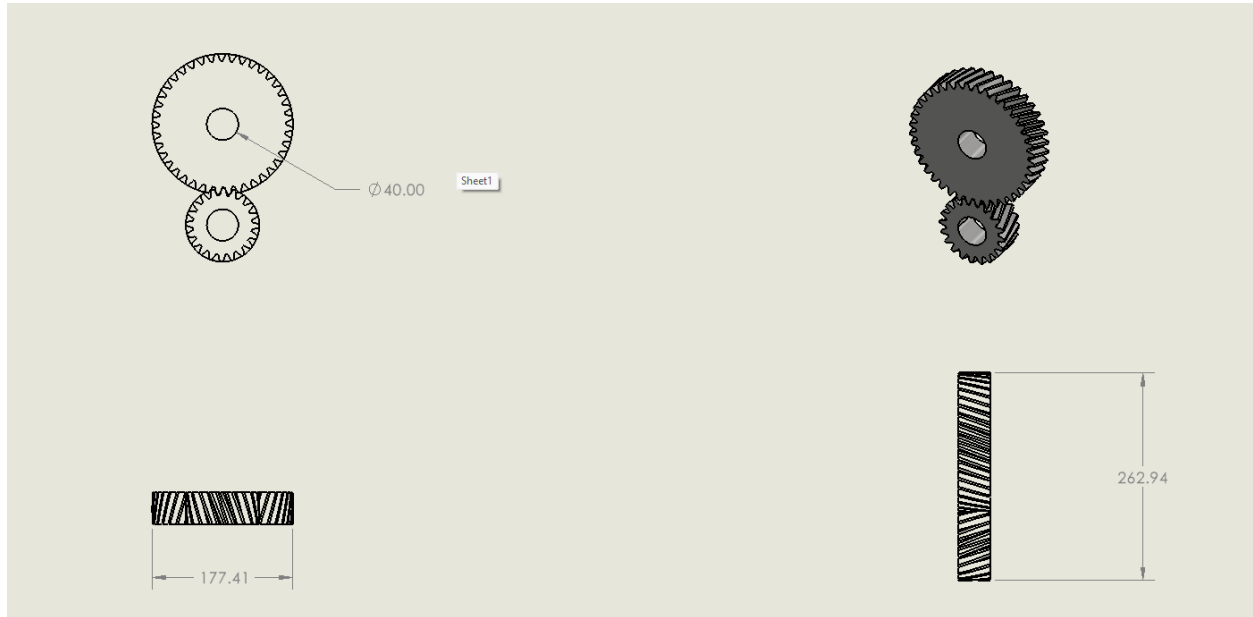
Mesh Independence Test: The final factor of safety was selected according to this test. Mesh elements were selected at around 1.2×10^5 .



Decision : Helical gear was selected because the factor of safety was found 1.3832 which falls into the range of 1.1-1.5 according to ISO 6336 and 1.2-1.5 according to AGMA 2101-D04.

5. Develop a Prototype

5.0.1 Detailed Drawing of gear



5.0.2 Manufacturing of Helical Gear

1. Material Selection and Preparation:

- **Material:** AISI 8620 is selected due to its excellent toughness, fatigue strength, and wear resistance.
- **Raw Material Preparation:** Cut AISI 8620 steel into appropriate-sized blanks using a band saw or laser cutter. Ensure the material is free from surface defects.

2. Forging (Optional):

- For large gears or applications needing superior grain flow, forge the material to improve mechanical properties.
- Normalize the forged blank to relieve internal stresses.

3. Preliminary Machining:

- Use a lathe to turn the blank to the required outer diameter, bore diameter, and face width.
- Machine keyways, splines, or other preliminary features as needed.

4. Gear Cutting:

- **Hobbing:** Mount the blank on a hobbing machine. Use a helical hob with the appropriate lead angle to cut the teeth progressively.
- **Milling (Alternative):** For smaller batches, use a CNC milling machine with a specialized helical gear cutter.
- Maintain tight tolerances to ensure minimal post-processing.

5. Heat Treatment:

- **Carburizing:** Heat the gear to around 930–950°C in a carbon-rich atmosphere to increase surface carbon content while keeping the core tough.
- **Quenching:** Rapidly cool the gear in oil or gas to harden the outer surface.
- **Tempering:** Reheat the gear to 150–200°C to reduce brittleness and improve toughness.

6. Finish Machining:

- **Grinding:** Perform gear tooth grinding to achieve high dimensional accuracy and surface finish, typically using a profile or continuous generation grinding process.
- **Honing (Optional):** For extremely tight tolerances, apply honing to further refine the gear tooth surfaces.

7. Inspection and Quality Control:

- Check dimensions, tooth profile, pitch, and lead with precision instruments like a coordinate measuring machine (CMM) or gear measuring center.
- Perform magnetic particle inspection (MPI) or ultrasonic testing to detect any internal flaws caused by heat treatment.

8. Surface Treatment:

- Apply protective coatings, such as phosphate coatings or anti-rust oil, to prevent corrosion.

9. Final Assembly and Testing:

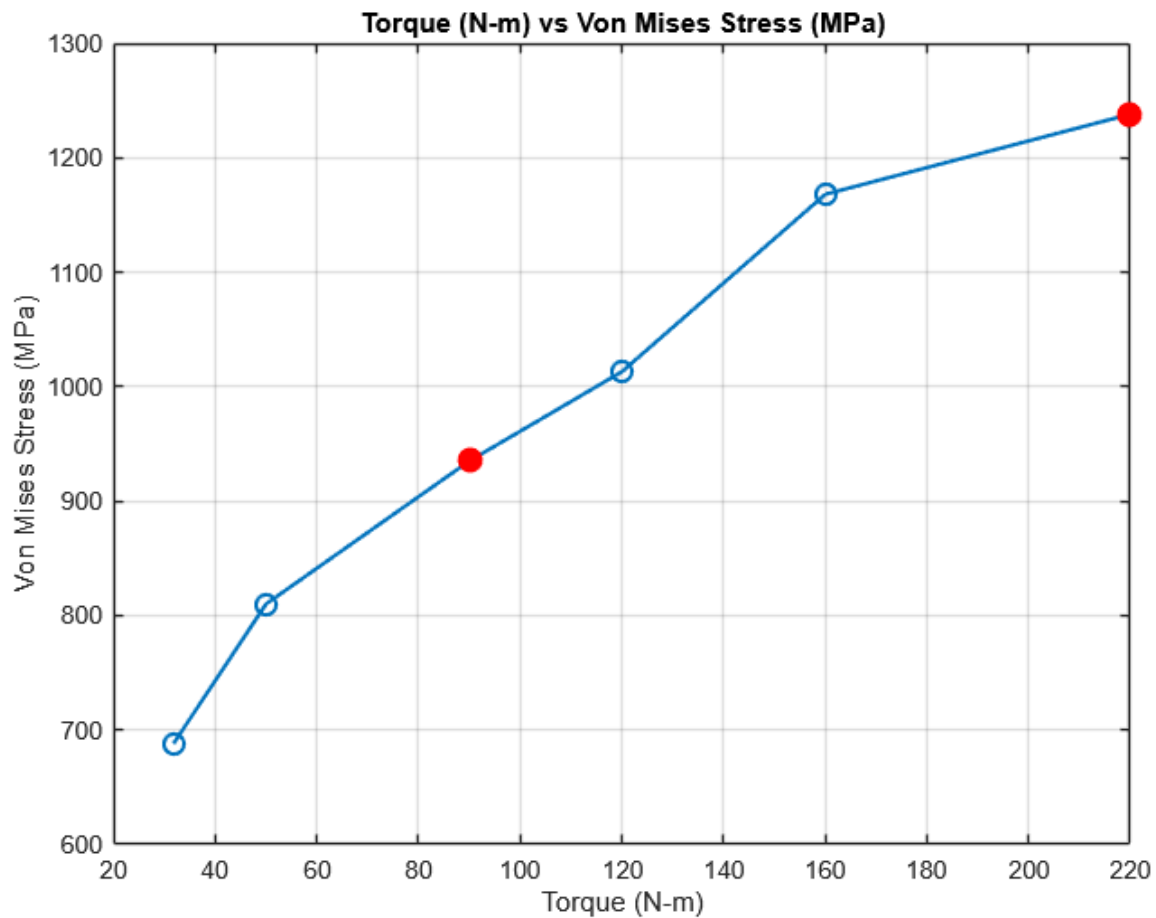
- Assemble the gear into the intended system, ensuring proper alignment and smooth meshing with mating gears.
- Conduct load testing and vibration analysis to verify performance.

10. Packaging and Delivery:

- Package the finished gears with corrosion inhibitors and ensure they are protected from mechanical damage during transportation.

6. Test and Evaluate

The gear was tested by increasing the applied torque and was plotted against AISI 8620 properties.



The red points indicate the yield strength and ultimate strength of the material respectively which are 910 MPa and 1240 MPa.

7. Redesign the Solution

Since the design factor of safety was within the range, there is no need for redesign.

8. Communicate Your Achievements

Future Improvements:

Lightweight Materials: Investigating innovative materials such as composite gears to decrease weight and boost efficiency.

Automated Lubrication System: Introducing a smart lubrication system to increase durability.

Noise Reduction Optimization: Continuously improving gear profiles to lessen vibrations and operational noise.

9. References

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- [14] A. G. Maitra, *Handbook of Gear Design*, 2nd ed. New Delhi, India: Tata McGraw-Hill, 2009.
- [15] ANSI/AGMA 2001-D04, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, 2004.

A Appendix A: Additional Calculations

Spur gear:

Spur Gears Component Generator

Design Calculation

Method of Strength Calculation
ISO 6336:1996

Loads

	Gear 1	Gear 2
Power	P 10 kW	9.800 kW
Speed	n 18840.00 rpm	9420.00 rpm
Torque	T 5.069 N m	9.935 N m
Efficiency	η 0.980 ul	

Material Values

Gear 1 ☒ A519-1050

Gear 2 ☒ A519-1050

Bending Fatigue Limit	σ_{Flim} 336.0 MPa	336.0 MPa
Contact Fatigue Limit	σ_{Hlim} 380.0 MPa	380.0 MPa
Modulus of Elasticity	E 206000 MPa	206000 MPa
Poisson's Ratio	μ 0.300 ul	0.300 ul
Heat Treatment	0 ul	0 ul

Required Life L_h 1000000 hr

Factors Accuracy

5:30 PM Calculation: Speed is higher than minimal critical speed
5:30 PM Design: Gear 1: The Unit Correction (x) is less than the Unit Correction without Tapering (x_t)
5:30 PM Design: Numbers of teeth are commensurable - shots of the same teeth are taken relatively re
5:30 PM Calculation: Calculation indicates design compliance!

Results

F_t	119.074 N
F_r	46.121 N
F_a	43.339 N
F_n	134.848 N
v	83.982 mps
n_{E1}	14722.382 rpm
Gear 1	
S_H	1.527 ul
S_F	21.539 ul
S_{Hst}	3.441 ul
S_{Fst}	47.015 ul
Gear 2	
S_H	1.527 ul
S_F	23.006 ul
S_{Hst}	3.441 ul
S_{Fst}	49.885 ul

Calculate OK Cancel <<

Type of Load Calculation
☒ Power, Speed --> Torque
☐ Torque, Speed --> Power
☐ Power, Torque --> Speed

Type of Strength Calculation
Check Calculation

Limit Values

	Contact	Bending
Minimal Factor of Safety	1.200 ul	1.300 ul

Helical gear:

Spur Gears Component Generator

Design

Calculation

Method of Strength Calculation

ISO 6336:1996

Loads

		Gear 1	Gear 2
Power	P	10.000 kW	9.800 kW
Speed	n	18840.00 rpm	9420.00 rpm
Torque	T	5.069 N m	9.935 N m
Efficiency	η	0.980 ul	

Material Values

Gear 1	☒	A519-1050	
Gear 2	☒	A519-1050	
Bending Fatigue Limit	σ_{Flim}	336.0 MPa	336.0 MPa
Contact Fatigue Limit	σ_{Hlim}	380.0 MPa	380.0 MPa
Modulus of Elasticity	E	206000 MPa	206000 MPa
Poisson's Ratio	μ	0.300 ul	0.300 ul
Heat Treatment		0 ul	0 ul

Required Life

L_h 100000 hr

Factors

Accuracy

3:04:30 PM Calculation: Speed is higher than minimal critical speed

3:04:30 PM Design: Gear 1: The Unit Correction (x) is less than the Unit Correction without Tapering (>)

3:04:30 PM Design: Numbers of teeth are commensurable - shots of the same teeth are taken relative

3:04:30 PM Calculation: Calculation indicates design compliance!

Results

F_t	126.716 N
F_r	46.121 N
F_a	0.000 N
F_n	134.848 N
v	78.917 mps
n_{E1}	16216.938 rpm
Gear 1	
S_H	1.437 ul
S_F	22.944 ul
S_{Hst}	3.348 ul
S_{Fst}	50.147 ul
Gear 2	
S_H	1.527 ul
S_F	25.011 ul
S_{Hst}	3.557 ul
S_{Fst}	54.332 ul

Calculate

OK

Cancel

<<

Type of Load Calculation

☒ Power, Speed --> Torque
☐ Torque, Speed --> Power
☐ Power, Torque --> Speed

Type of Strength Calculation

Check Calculation

Limit Values

	Contact	Bending
Minimal Factor of Safety	1.200 ul	1.300 ul